



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

Program: BTech (CSE) Stream: CyberSecurity

Subject: Application Security Testing

Date: 27/June/2023

Marks: 100

Year: III Semester: VI

Time: 3 hrs (2 to 5pm)

No. of Pages: 02

Re-Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	
CO-1; SO-1; BL-2; U1	a.	How is motive, goal and objective related to Information Security Attacks and comment on any TWO Information Security Attack Vectors.	[05]
CO-2; SO-1; BL-1; U2	b.	Describe the BSIMM Framework for assessing the Secure Software Lifecycle.	[05]
CO-3; SO-4; BL-2; U3	c.	Jot down the reasons for Memory Management Vulnerability.	[05]
CO-2; SO-1; BL-2; U5	d.	Infer and describe the technique for taking over a mobile device via Bluetooth.	[05]
Q2 CO-1; SO-1; BL-3; U1	a.	Working and Significance of MVC Architecture compared with Micro Services Architecture.	[10]
Q2 CO-2; SO-1; BL-2; U4	b.	Specify three attack techniques of HTTP Request Smuggling with example.	[10]

Q3 CO-2; SO-1; BL-4; U3	a.	Comment on Dynamic Detection techniques and relate to how and which vulnerability it helps in detecting.	[10]
Q3 CO-3; SO-4; BL-4; U4	b.	Difference between Server Side Injection and Cross Site Scripting attacks wrt their working.	[10]
Q4 CO-1; SO-1; BL-2; U2	a.	Infer the need for adaptation of SecSDL to cloud environment.	[10]
Q4 CO-2; SO-1; BL-1; U5	b.	Explain any FIVE OWASP Mobile Top10 Testing with an example for each.	[10]
Q5 CO-2; SO-1; BL-2; U3	a.	How would sandboxing help in mitigating race condition vulnerability and memory management vulnerability.	[10]
Q5 CO-3; SO-4; BL-1; U5	b.	Describe Mobile Attack Vectors/Vulnerabilities.	[10]
Q6 CO-1; SO-1; BL-4; U1	a.	Compare between WAF, RASP and CNAPP application security tools and solutions.	[10]
Q6 CO-2; SO-1; BL-2; U4	b.	Explain FIVE types of SQLi vulnerabilities with example.	[10]
Q7 CO-2; SO-1; BL-2; U2	a.	Explain the SDL practices outlined by SAFECODE to increase trust in information and communication technology.	[10]
Q7 CO-3; SO-4; BL-4; U4	b.	Comment on ways to mitigate Broken Authentication Vulnerability and differentiate between Single-factor and Multi-factor Authentication.	[10]